AVIOTHIC

Clinical AI Systems

Breast Cancer Report Copilot

Implementation Plan

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This document outlines the technical plan for an AI assistant that analyzes uploaded breast imaging and pathology reports, builds a structured patient context, and supports grounded clinical follow-up conversation. The system is designed to assist clinicians by organizing information and explaining findings — it does not replace clinical judgment.

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1. Product Vision

What the copilot does for the clinician, end to end.

- Upload pathology, mammography, ultrasound, MRI, PET-CT, or discharge reports.
- Extract structured clinical entities: BI-RADS, TNM, tumor size, biomarkers, lymph nodes, and recommendations.
- Create a patient-specific context memory.
- Allow radiologists to ask unlimited follow-up questions about the uploaded report.
- Retrieve trusted breast cancer knowledge when needed, via retrieval-augmented generation (RAG).
- Generate structured summaries and downloadable reports.

2. System Workflow

The pipeline a report follows from upload to conversation.

Step	Stage
1	Upload PDF / image
2	OCR (if needed)
3	Text cleaning
4	Medical entity extraction
5	LLM reasoning
6	RAG retrieval
7	Patient context memory
8	Conversational AI
9	Structured summary & JSON output

3. Recommended Tech Stack

Primary technology choices by function.

Function	Choice
Language	Python 3.11

Function	Choice
Editor	VS Code
Framework	FastAPI
LLM	Qwen 3 8B or Llama 3 8B
Fine-tuning	Unsloth + PEFT (QLoRA)
Embeddings	bge-large-en or BioClinicalBERT embeddings
Vector database	ChromaDB
OCR	Tesseract + PyMuPDF
PDF parsing	pdfplumber / PyMuPDF
NER	spaCy + custom medical rules
Inference	vLLM
Experiment tracking	Weights & Biases
Container	Docker
Deployment	RunPod / Vast.ai

4. Core Modules

The functional building blocks of the system.

- Document ingestion
- OCR pipeline
- Medical entity extraction
- Patient context builder
- Breast cancer knowledge base
- RAG engine
- Conversation engine
- Safety & citation layer
- Structured report generator

5. Datasets

Data sources for training, fine-tuning, and evaluation.

- CBIS-DDSM
- VinDr-Mammo
- PubMed abstracts
- PubMed Central Open Access
- Breast cancer guidelines, where licensing permits use
- Synthetic instruction datasets created from public sources

6. LLM Capabilities

What the model is expected to do once integrated.

- Summarize reports
- Extract biomarkers
- Explain BI-RADS
- Explain staging
- Highlight missing information
- Answer follow-up questions using report context
- Compare two reports
- Explain findings in simple language
- Generate structured JSON

7. Development Phases

Sequenced delivery plan from data to deployment.

Phase	Focus
Phase 1	Data ingestion and preprocessing
Phase 2	OCR and report parser
Phase 3	Entity extraction
Phase 4	Fine-tune oncology LLM
Phase 5	Build RAG
Phase 6	Build patient context memory
Phase 7	Conversational assistant
Phase 8	Evaluation and deployment

8. Evaluation

How quality, safety, and performance are measured before release.

- Extraction precision / recall
- Summary quality review
- Hallucination testing
- Latency
- Clinical review with radiologists
- Regression testing

9. Final Deliverables

What is handed over at the end of the engagement.

- Breast cancer report copilot
- Conversation interface
- Patient context engine
- RAG knowledge base
- Fine-tuned oncology assistant
- Technical documentation
- Deployment guide